

Food Component Detection for Dietary Recommendation-Children's Fun Nutrition Guide

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Abstract — This paper presents the design, implementation, and preliminary evaluation of the Children's Fun Nutrition Guide, a web-based platform aimed at improving dietary habits among children aged 3 to 12. Motivated by growing concerns about childhood obesity and poor nutritional outcomes documented by the WHO and CDC, the platform integrates an AI-powered nutritional assistant (Nutri), a recipe recommendation engine, and educational mini-games in a secure environment. Website: <https://kids-nutrition-fun.com/>

Keywords — *Children's Nutrition; AI chatbot; Recipe Recommendation; Gamification; Web Application*

I. INTRODUCTION

Children aged 3 to 12 represent a critical window for establishing lifelong dietary habits. According to the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC), poor nutrition during early childhood is strongly associated with developmental delays, obesity, weakened immunity, and increased risk of chronic diseases such as type-2 diabetes [1][2]. There is a clear gap in the market for a platform that combines evidence-based dietary recommendations with a child-friendly design. This project addresses that gap by developing the Children's Fun Nutrition Guide, a web-based platform designed to make healthy eating an engaging adventure for children aged 3-12 and their parents.

The platform leverages AI to deliver personalized recipe recommendations, integrates educational mini-games, and a user login system—all within a secure environment. This paper presents the design rationale, system architecture, core features, and preliminary evaluation of the platform.

II. DESIGN/METHODOLOGY/IMPLEMENTATION

The platform is built on WordPress with custom PHP and JavaScript extensions. The backend integrates the PolyU COMP OpenAI API gateway—a departmental Azure OpenAI proxy—to power the AI assistant. The deployment used is gpt4o-mini (GPT-4o Mini, version 2024-07-18), selected for its cost efficiency at \$0.00066 per 1,000 tokens [3]. User data is stored in MySQL with role-based access control.

A. Nutritional Assistant (Nutri)

Nutri is a bilingual (English/Chinese) chatbot embedded site-wide. Each user message is submitted to the Chat Completions API with a curated system prompt that restricts responses to child nutrition topics, ensuring age-appropriate and safe content. Token usage is tracked per the gateway's scaling factor to monitor budget consumption.

B. Recipe Recommendations

Parents can browse and access 50+ personalized, kid-friendly recipes tailored to their child's age, allergies, and health goals, with recommendations aligned to WHO and CDC dietary

guidelines. Each recipe includes detailed information on ingredients and nutritional content.

C. Educational Mini-Games

Two browser-based educational mini-games — Nutrition Parkour and Nutrition Plate Builder — reinforce children's nutritional literacy through interactive gameplay, helping kids aged 3–12 identify healthy foods and understand balanced meal composition in an engaging, age-appropriate format. Both games are embedded via iframe and are fully playable on desktop and mobile.

III. EVALUATION AND RESULTS

We conducted an initial usability test with 20 parents (including teachers and teenagers) of children aged 3-12. Key findings include:

- 95% of testers want to recommend this website to other parents, teachers, or caregivers
- 100% of users find the AI nutrition assistant "Nutri" very helpful.
- Participants gave the platform an average score of 4.9 out of 5 for its visual design.

The AI assistant successfully responded to over 95% of nutrition-related queries in testing, with appropriate bilingual responses.

IV. CONCLUSION

The Children's Fun Nutrition Guide demonstrates that AI-assisted personalization and gamification can effectively bridge the gap between clinical nutrition guidelines and practical adoption in families. By integrating the PolyU COMP OpenAI API gateway with a child-safe, bilingual conversational assistant, recipe recommendations, and interactive games, the platform delivers a holistic solution to childhood nutrition challenges for ages 3–12. Future work will incorporate image-based food logging and longitudinal behavioral studies.

REFERENCES

- [1] World Health Organization, "Healthy diet," WHO Fact Sheet, Apr. 2020. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>
- [2] Centers for Disease Control and Prevention, "Child Nutrition," CDC, 2023. [Online]. Available: <https://www.cdc.gov/nutrition/childhood-nutrition>
- [3] The Hong Kong Polytechnic University COMP Dept., "COMP OpenAI Access Guide," Ver. 3.6, Aug. 2025.